

Unit ID: 6798d0f3aaf593b2823f180a

Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738068209335-Chapter 7_unit 2 - Bringing Ideas to Life.pdf

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UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

TIME WELL MANAGED: CATHY'S COUNTDOWN CREATION

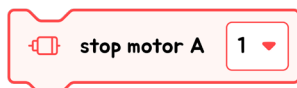
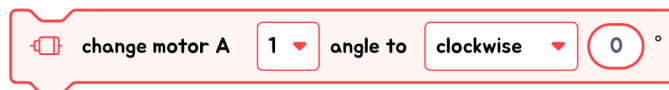


In the bustling halls of Lux Middle School, Cathy, a thoughtful and inventive seventh grader, noticed that many of her classmates struggled to manage their time efficiently during exams. They often mentioned how quickly time seemed to pass, leaving some questions unanswered. Inspired by this common problem, Cathy decided to create a practical solution for her next school project—a Timer with Dial Control. She envisioned a timer that students could easily adjust to reflect the duration of their exams or study sessions, making it more interactive and user-friendly than distracting digital apps.

Gathering her materials, including a MODI module kit, Cathy began designing her timer in the school's makerspace. She programmed the device so that turning the dial would set the countdown, allowing users to visually track how much time they had left. The timer also featured an alert system that beeped gently at the halfway point and emitted quicker beeps during the last minute to help students manage their pace. As she refined her design based on feedback from intrigued classmates and teachers, Cathy prepared to debut her timer at the school science fair. Her creation not only helped her peers manage their time better but also sparked discussions on the importance of pacing and time awareness in academic success.



ABOUT MOTOR A MODULE

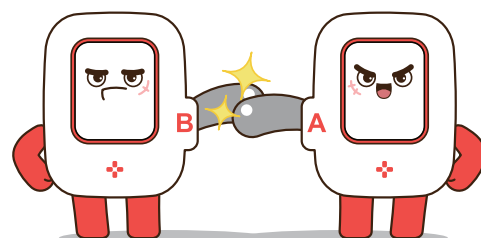
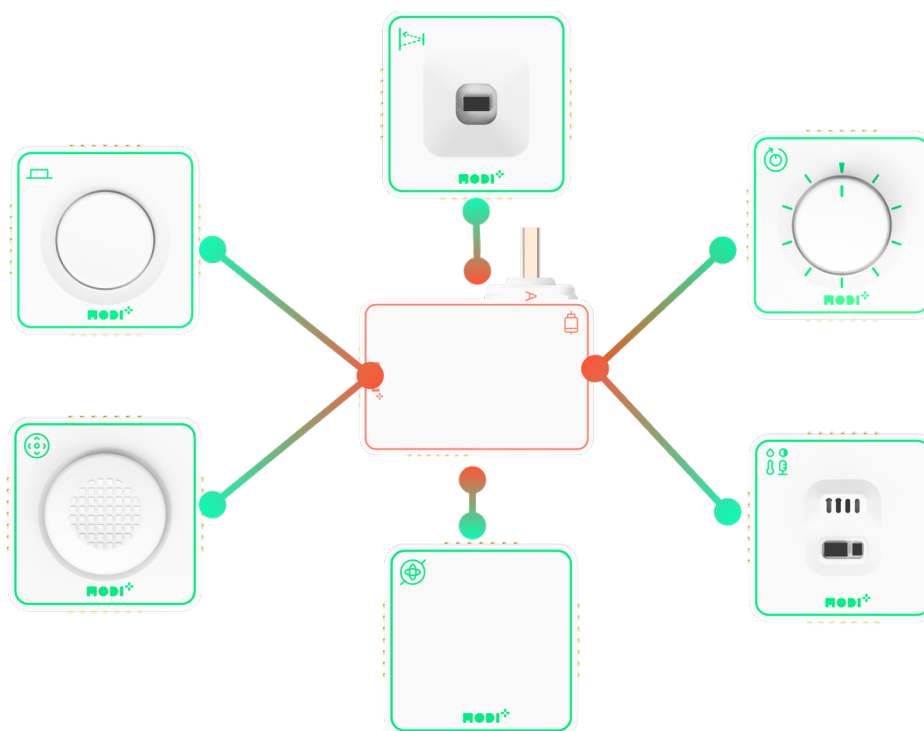


Motor A features five different blocks in the block area. These blocks allow you to customize your creations by setting the motor's angle, spinning direction, and speed.

INTERACTIONS WITH OTHER MODULES

As an output module, Motor A can interact with all input modules, showing a unique response to each one. Explore these interactions by connecting Motor A to different input modules!

INTERACTION BETWEEN LED AND OTHER MODULES





Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?



Dial



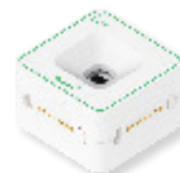
Button



Environment



Joystick



ToF



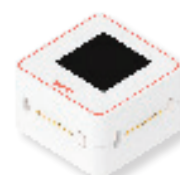
Speaker



IMU



LED



Display

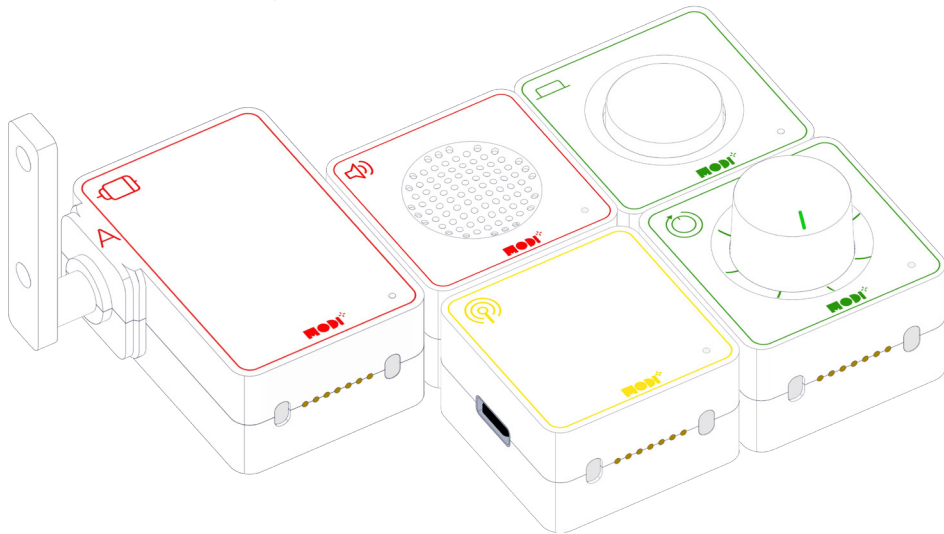
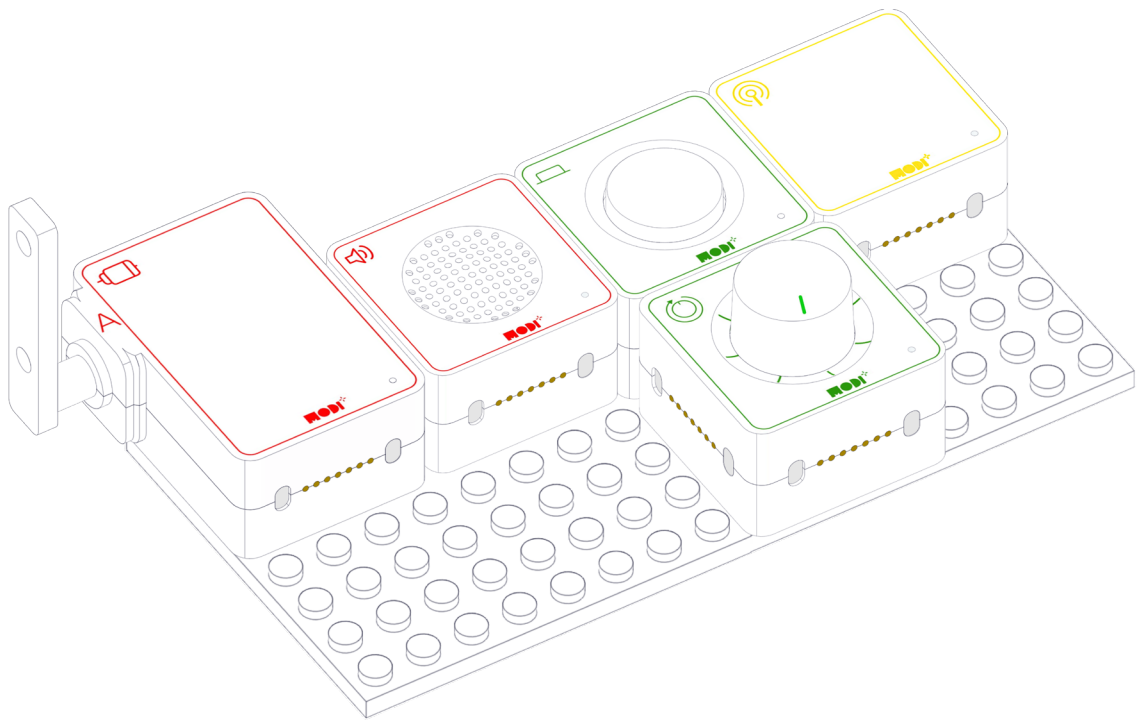


Motor A



Motor B

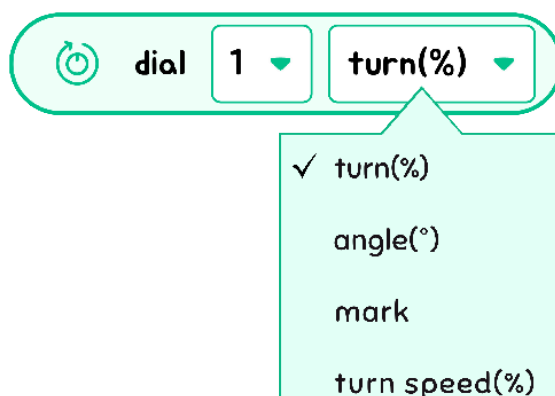
HOW DOES THE CREATION LOOK LIKE?





Module Functionality

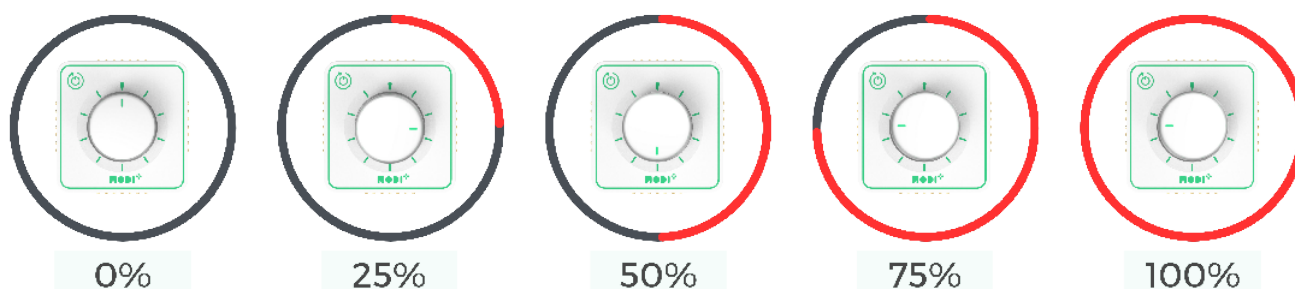
Let's look detail about the Dial module.



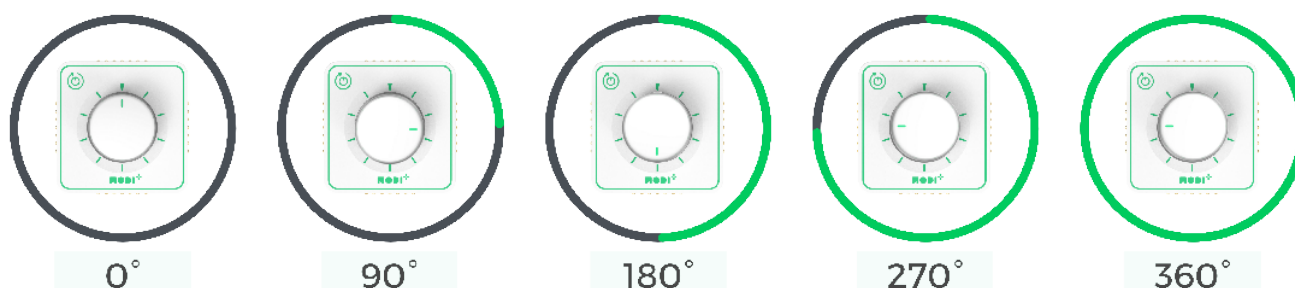
FUNCTIONS IN DETAIL

1. Turn (%) - Range: 0% ~ 100% This just means moving the dial around, either to the right or left.	2. Angle (°) - Range: 0~360° This is how far you've moved the dial from the default (0°), measured in degrees.
3. Mark - Range: 1~10 Marks on the dial show where it can be turned to, like the numbers on a clock.	4. Turn Speed (%) - Range: 0% ~ 100% This tells us how fast you are turning the dial, in percentage.

TURN (%)



ANGLE (°)



The Dial module can rotate from 0 to 360 degrees, similar to turning a knob full circle. For simplicity, we'll consider a full rotation as 100%. Therefore, a half-turn, or 180 degrees, represents 50% of a full rotation.